

JavaScript Web Storage API

[Back to JS page](#)

Table of Contents

- [JavaScript Web Storage API](#)
 - [The localStorage Object](#)
 - [Example 1](#)
 - [The setItem\(\) and getItem\(\) Methods](#)
 - [Example 2](#)
 - [The sessionStorage Object](#)
 - [Example 3](#)
 - [Storage Object Properties and Methods](#)
 - [Example 4](#)
 - [Practical Example - Visitor Counter](#)
 - [Example 5](#)
 - [Document](#)
 - [Reference](#)

The localStorage Object

The `localStorage` object stores data with no expiration date. The data is NOT deleted when the browser is closed, and will be available the next day, week, or year.

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Web Storage API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The localStorage Object</h4>
<p id="demo"></p>

<script>
// Store data
localStorage.setItem("name", "John Doe");
localStorage.setItem("age", "30");
localStorage.setItem("city", "New York");

// Retrieve data
document.getElementById("demo").innerHTML =
  "Stored in localStorage:<br><br>" +
  "name: " + localStorage.getItem("name") + "<br>" +
  "age: " + localStorage.getItem("age") + "<br>" +
  "city: " + localStorage.getItem("city");
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

The setItem() and getItem() Methods

`setItem(key, value)` stores data. `getItem(key)` retrieves data. Values are stored as strings:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Web Storage API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The setItem() and getItem() Methods</h4>

<input type="text" id="keyInput" placeholder="Key" value="color">
<input type="text" id="valueInput" placeholder="Value" value="blue">
<br><br>
<button onclick="saveItem()">Save Item</button>
<button onclick="loadItem()">Load Item</button>
<button onclick="removeItem()">Remove Item</button>
<p id="demo2"></p>

<script>
function saveItem() {
    const key = document.getElementById("keyInput").value;
    const value = document.getElementById("valueInput").value;
    localStorage.setItem(key, value);
    document.getElementById("demo2").innerHTML =
        "Saved: " + key + " = " + value;
}

function loadItem() {
    const key = document.getElementById("keyInput").value;
    const value = localStorage.getItem(key);
    if (value !== null) {
        document.getElementById("demo2").innerHTML =
            "Loaded: " + key + " = " + value;
    } else {
        document.getElementById("demo2").innerHTML =
            "Key '" + key + "' not found!";
    }
}

function removeItem() {
    const key = document.getElementById("keyInput").value;
    localStorage.removeItem(key);
    document.getElementById("demo2").innerHTML =
        "Removed: " + key;
}
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

The sessionStorage Object

The `sessionStorage` object stores data for one session only. The data is deleted when the browser tab is closed:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Web Storage API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>The sessionStorage Object</h4>
<p id="demo3"></p>

<script>
// Store session data
sessionStorage.setItem("sessionId", "abc123");
sessionStorage.setItem("startTime", new Date().toLocaleTimeString());

// Retrieve session data
document.getElementById("demo3").innerHTML =
  "sessionStorage data (will be deleted when tab closes):<br><br>" +
  "sessionId: " + sessionStorage.getItem("sessionId") + "<br>" +
  "startTime: " + sessionStorage.getItem("startTime");
</script>

</body>
</html>
```

Example 3

Result [View Example](#)

Storage Object Properties and Methods

The Storage object provides these properties and methods:

Method/Property	Description
<code>setItem(key, value)</code>	Stores a key/value pair
<code>getItem(key)</code>	Retrieves a value by key
<code>removeItem(key)</code>	Removes a key/value pair
<code>clear()</code>	Removes all key/value pairs
<code>key(index)</code>	Gets the key at a specific index
<code>length</code>	Returns the number of stored items

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Web Storage API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Storage Object Properties and Methods</h4>
<p id="demo4"></p>

<button onclick="showAll()">Show All Items</button>
<button onclick="clearAll()">Clear All</button>

<script>
function showAll() {
    let text = "Number of items: " + localStorage.length + "<br><br>";

    for (let i = 0; i < localStorage.length; i++) {
        const key = localStorage.key(i);
        const value = localStorage.getItem(key);
        text += "[" + i + "] " + key + ": " + value + "<br>";
    }

    if (localStorage.length === 0) {
        text += "(empty)";
    }

    document.getElementById("demo4").innerHTML = text;
}

function clearAll() {
    localStorage.clear();
    document.getElementById("demo4").innerHTML = "All items cleared! (" + localStorage.length + "
}

// Add some demo items if empty
if (localStorage.length === 0) {
    localStorage.setItem("demo1", "value1");
    localStorage.setItem("demo2", "value2");
    localStorage.setItem("demo3", "value3");
}
</script>

</body>
</html>
```

Example 4

Result [View Example](#)

Practical Example - Visitor Counter

A practical example using localStorage to count and remember visitors:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Web Storage API</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Practical Example – Visitor Counter</h4>
<p id="counter"></p>
<p id="firstVisit"></p>
<p id="lastVisit"></p>

<button onclick="resetCounter()">Reset Counter</button>

<script>
function updateCounter() {
    let count = localStorage.getItem("visitCount");

    if (count === null) {
        count = 0;
        localStorage.setItem("firstVisit", new Date().toLocaleString());
    } else {
        count = parseInt(count);
    }

    count++;
    localStorage.setItem("visitCount", count);
    localStorage.setItem("lastVisit", new Date().toLocaleString());

    document.getElementById("counter").innerHTML =
        "You have visited this page " + count + " time(s).";

    document.getElementById("firstVisit").innerHTML =
        "First visit: " + localStorage.getItem("firstVisit");

    document.getElementById("lastVisit").innerHTML =
        "Last visit: " + localStorage.getItem("lastVisit");
}

function resetCounter() {
    localStorage.removeItem("visitCount");
    localStorage.removeItem("firstVisit");
    localStorage.removeItem("lastVisit");
    updateCounter();
}

updateCounter();
</script>
```



```
</body>  
</html>
```

Example 5

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript Web Storage API](#)